Computational Physics: Numerical Simulation of Planetary Orbits

1. Gravitational Dynamics and Differential Equations

Newtonian gravitational acceleration vector between mass M and test particle satisfies:

$$\mathbf{a} = -\frac{GM}{r^3} \mathbf{r}$$

2. Verlet Integration Algorithm in Python

```
import numpy as np

def verlet_step(pos: np.ndarray, vel: np.ndarray, dt: float, G: float = 6.674e-11, M: float = 1.989e30):
    """Symplectic numerical integrator preserving orbital energy."""
    r = np.linalg.norm(pos)
    acc = - (G * M / (r**3)) * pos
    new_pos = pos + vel * dt + 0.5 * acc * (dt**2)
    new_r = np.linalg.norm(new_pos)
    new_acc = - (G * M / (new_r**3)) * new_pos
    new_vel = vel + 0.5 * (acc + new_acc) * dt
    return new_pos, new_vel
```

Table 1.1: Error Convergence of Numerical Orbit Integrators

Integrator	Order	Energy Drift (100 Orbits)	Symplectic
Euler-Forward	O(dt)	+14.82%	No
Runge-Kutta 4	O(dt^4)	+0.004%	No
Verlet-Velocity	O(dt^2)	< 0.0001%	Yes